ACCURATE FAIRNESS

One loan application. Two similar applicants. Four ways the model can decide.

TWO SIMILAR APPLICANTS...



SIMILAR



(x, a) (x, a')

Same income, same credit history, same job: the same x

Only the sensitive attribute differs (gender): a vs a'

Ground TRUTH: application should be approved (y)

Accurate fairness: both decisions must match y

$$D(y, f(x,a')) \leq K \cdot d((x,a), (x,a'))$$

ACCURATE FAIR (AF)

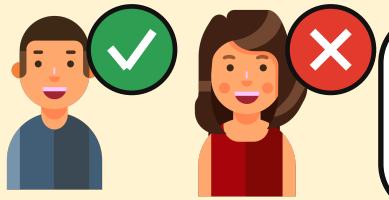


Both approved — and the truth says approve. Correct AND the same for both!

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ACCURATE AND FAIR — THIS IS THE GOAL

ACCURATE BIAS (AB)

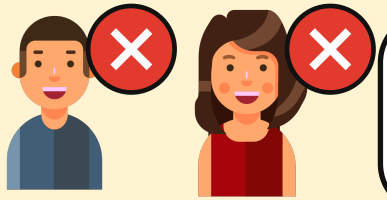


He's approved, correctly. But I'm rejected?! The decision depended on gender, not merit.

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ACCURATE BUT UNFAIR — accuracy-only metric miss this

INACCURATE FAIR (IF)

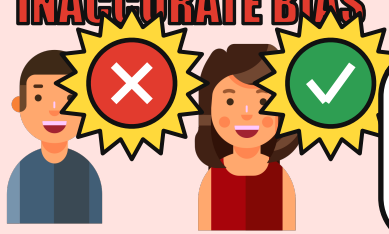


We're treated the same... but we're BOTH wrongly rejected. Fair by being wrong!

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FAIR BUT INACCURATE — fairness-only metric miss this

INACCURATE BIAS (IB)



He's wrongly rejected, and I get a different answer. Nothing went right.

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NEITHER ACCURATE NOR FAIR — the worst case

INDIVIDUAL PREDICTED CORRECTLY

INDIVIDUAL PREDICTED WRONGLY